Experiment:5 Electromagnetic Simulation and S-Parameter Characterization of a Microwave Directional Coupler and Two-Way Power Divider

Aim: To simulate a microwave directional coupler and a two-way power divider using MATLAB and analyze their S-parameters.

Apparatus:

Computer/Laptop
MATLAB software

Procedure:

- Open MATLAB and create a new script.
- Enter the code for the directional coupler and two-way power divider.
- Define the frequency range and S-parameter values.
- Plot the S-parameters using subplot() to display three figures in one window.
- Run the program and observe the three plots.
- Compare the transmission characteristics of both devices.

Code:

```
clc;

clear;

close all;

f = linspace(1,10,500);

S11c = 0.1*ones(size(f));

S21c = 0.7*ones(size(f));

S31c = 0.7*ones(size(f));

S41c = 0.05*ones(size(f));

S11d = 0.05*ones(size(f));

S21d = 1/sqrt(2)*ones(size(f));

S31d = 1/sqrt(2)*ones(size(f));

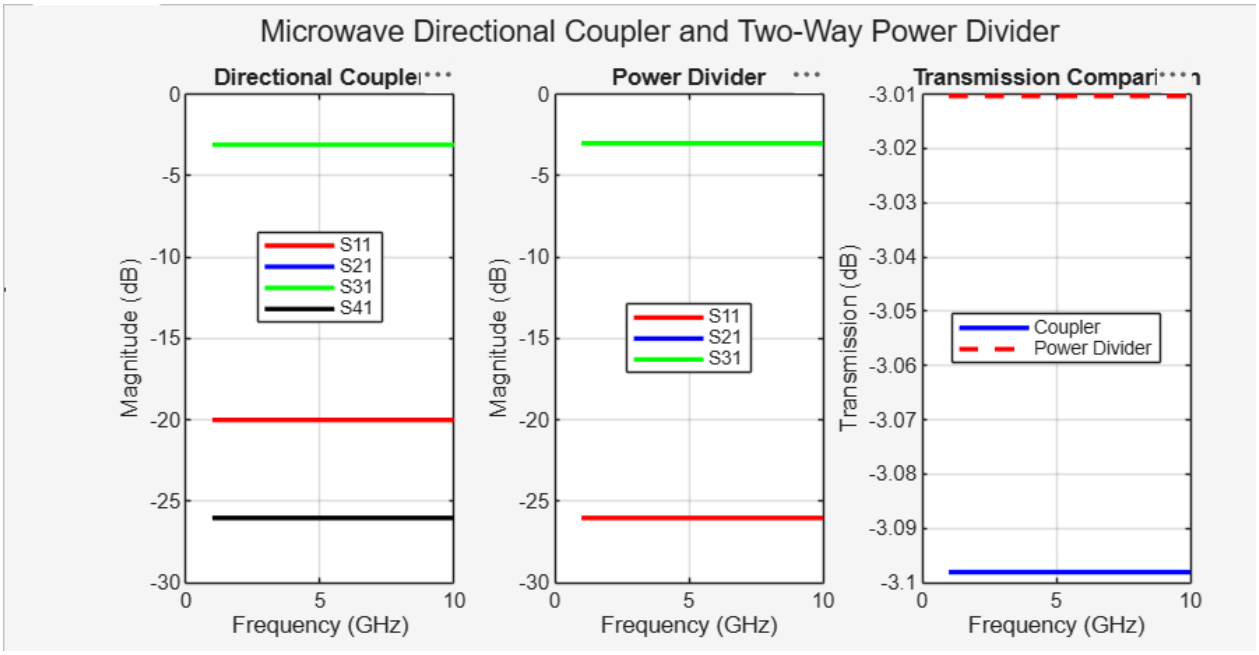

figure;
```

```
subplot(1,3,1);
plot(f,20*log10(S11c),'r','LineWidth',2);
hold on;
plot(f,20*log10(S21c),'b','LineWidth',2);
plot(f,20*log10(S31c),'g','LineWidth',2);
plot(f,20*log10(S41c),'k','LineWidth',2);
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Directional Coupler');
legend('S11','S21','S31','S41','Location','best');
subplot(1,3,2);
plot(f,20*log10(S11d),'r','LineWidth',2);
hold on;
plot(f,20*log10(S21d),'b','LineWidth',2);
plot(f,20*log10(S31d),'g','LineWidth',2);
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Power Divider');
legend('S11','S21','S31','Location','best');

subplot(1,3,3);
plot(f,20*log10(S21c),'b','LineWidth',2);
hold on;
plot(f,20*log10(S21d),'r--','LineWidth',2);
grid on;
```

```
xlabel('Frequency (GHz)');
ylabel('Transmission (dB)');
title('Transmission Comparison');
legend('Coupler','Power Divider','Location','best');
sgtitle('Microwave Directional Coupler and Two-Way Power Divider');
```

Output:



Result:

The microwave directional coupler and two-way power divider were simulated using MATLAB, and their S-parameters and transmission characteristics were plotted and compared successfully